

1. Submission status

v3.6.3-guarded-seed-stage1-fill is a separate upload candidate. It retains the v3.5 Stage-A model, hybrid random-forest family router, largest-component ROI policy, anatomy-specific Stage-B experts, and all neural-network weights. The v3.6 A1 decoder and v3.6.1 split-candidate RF are disabled. The validated deterministic guarded-seed refinement and Stage-1 instance-support fill are enabled.

2. Method overview

The submission performs per-fragment instance segmentation of pelvic and femoral fractures with a serial two-stage STU-Net-B cascade:

- 1. **Stage A — anatomy.** A 5-class whole-volume model identifies sacrum, left/right hipbone and femur.
- 2. **Hybrid family routing.** A random forest selects pelvic or femur when confident; the organizers' acquisition rule is the uncertainty tiebreak.
- 3. **Stage B — fracture experts.** The routed anatomy selects a Sacrum, shared-Left/Right-Hip or Femur expert. It emits four ABBC channels and nine learned same-instance affinity channels at 1-, 3- and 9-voxel offsets.
- 4. **Safe base.** The exact v3.5 one-voxel average-linkage affinity decoder at $T=0.75$ produces the initial instance partition.
- 5. **Guarded seed recovery.** Within each base instance, ABBC Boundary evidence proposes a binary split. It is committed only when each final piece is at least 1,000 mm³, each contains at least 50 Core voxels, and all three affinity ranges provide separating evidence. At most two refinement passes are run. Post-split hard merging is disabled.
- 6. **Stage-1 support fill.** Within each 26-connected Stage-1 anatomy component, empty Stage-1 voxels are assigned to the nearest existing Stage-2 instance by watershed. Existing Stage-2 support and other anatomies are never overwritten. A component with no Stage-2 marker remains empty.
- 7. **Assembly.** Instances use the official ranges: sacrum 1-50, left hip 51-100, right hip 101-150 and femur 151-200.

3. Frozen deployment contract

Component	Deployed value
Stage A	Dataset539_PelvicFemurAnatomyV3, PenguinTrainerSTUNetBaseAnatomyV301, fold 0
Family	v3.3 hybrid RF, confidence margin 0.15

Component	Deployed value
router	
Stage B dataset	Dataset538_PelvicFemurBICMFragmentV5
Stage B experts	...V308SacrumExpertDeployedVal, ...V308HipExpertDeployedVal, ..V308FemurExpertDeployedVal
Stage B fold/checkpoint	fold 0 / checkpoint_best.pth
Stage B output	13 channels: 4 ABBC + 9 affinity
Safe base	v3.5 one-voxel affinity agglomeration, T=0.75
Split passes	2
Minimum final piece	1,000 mm ³
Minimum Core support	50 voxels on each side
Affinity agreement	separating evidence from 3/3 ranges: 1, 3 and 9 voxels
Post-split hard merge	disabled
Stage-1 fill	enabled after guarded seed; 26-connectivity; no distance cap

The model archive is byte-identical to v3.5. No new learned parameters or RF artifact are added for v3.6.3.

4. Model archive layout

Grand Challenge extracts the archive directly under `/opt/ml/model`:

```
/opt/ml/model/
```

```

└─ nnunet/results/

|   └─ Dataset539_PelvicFemurAnatomyV3/

|   |   └─ PenguinTrainerSTUNetBaseAnatomyV301___.../fold_0/

|   └─ Dataset538_PelvicFemurBICMFragmentV5/

|       └─ PenguinTrainerSTUNetBaseAffinityV308SacrumExpertDeployedVal___.../fold_0/

|       └─ PenguinTrainerSTUNetBaseAffinityV308HipExpertDeployedVal___.../fold_0/

|       └─ PenguinTrainerSTUNetBaseAffinityV308FemurExpertDeployedVal___.../fold_0/

└─ stage1_router/stage1_target_router_fold0.joblib

```

5. Local evaluation

The frozen evaluation contains 68 patients and 132 anatomy samples and uses the official-aligned local proxy. It is not a hidden-test leaderboard score.

De co de r	Dic e	HD 95 m m	AS SD m m	Re call	Pre cisi on
v3. 5 saf e ba se	0.8 556 51	6.6 874 90	1.8 522 38	0.9 314 12	0.9 32 19 7
v3. 6.2 gu ard ed se ed	0.8 627 38	6.3 586 05	1.7 656 82	0.9 37 72 5	0.9 309 34
v3	0.8	6.2	1.7	0.9	0.9

De co de r	Dic e	HD 95 m m	AS SD m m	Re call	Pre cisi on
.6.					
3					
+					
St	65	61	20	366	309
ag	62	98	75	43	34
e-	4	5	3		
1					
fill					

The combined method improves overlap, both surface distances, Recall, Instance F1 and Merge over v3.5. Relative to v3.6.2 guarded seed alone, it improves Dice/HD95/ASSD and Merge, while F1 is slightly lower. Recall for 1-5 cm³ GT fragments changes from 32/53 to 31/53; recall for fragments at least 5 cm³ remains 259/267. Split increases from 337 to 340.

6. Reproducibility and limitations

- The guarded-seed parameters above are the exact configuration evaluated on the frozen local arrays.
- Runtime checks fail fast if fewer than nine affinity channels are present, if any hard merge is accepted, or if Stage-1 fill changes the instance count.
- Fill can only add support inside the routed post-largest-component Stage-1 mask; it does not clip existing Stage-2 support or create marker-free objects.
- The local proxy was used to compare decoder variants, so its results may be optimistic. Hidden-test generalization must be established by submission.
- Stage-1 support fill has a measured trade-off: it reduces 1-5 cm³ fragment recall by one object and adds three split errors despite improving geometry.
- The 10-minute platform runtime and non-root execution must be checked before activation.

7. Runtime and licensing

The image targets an NVIDIA T4 16 GiB GPU. Stage A and each Stage-B expert are loaded serially; guarded-seed refinement and Stage-1 fill are deterministic CPU post-processing. STU-Net-B and nnU-Net are used under their respective open-source licenses; Stage-B initialization derives from the official TotalSegmentator base_ep4k checkpoint.

8. References

- Isensee, F. et al. nnU-Net, *Nature Methods* 18, 203-211 (2021).
- STU-Net / TotalSegmentator bone pretraining.
- PENGWIN 2026 Task 1 baseline:

github.com/YzzLiu/PENGWIN2026_Task1_AutoSeg_Baseline.